SIMATS Engineering • CSE • CSA0804 — Python Programming • 24-07-2026 • Category: Analytical • Student Submissions Report

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Q1. Harshad number

ANSWER

```
n=int(input())
temp=n
sum=0
while temp>0:
        digit=temp%10
        sum=sum+ digit
        temp=temp//10
if n% sum==0:
    print(n, "is a Harshad Number")
else:
    print(n, "is not a Harshad Number")
print("Harshad Numbers from 1 to 500:")
for i in range(1,501):
     temp=i
     s=0
     while temp>0:
             digit=temp%10
             s=s+ digit
             temp=temp//10
    if i% s == 0:
        print(i, end=" ")
```

Q2. GCD

CODE

```
a=int(input("Enter first:"))
b=int(input("Enter second:"))

while b ≠ 0:
    temp = a
    a=b
    b= temp % b
print("GCD is",a)
```

INPUT

```
24
36
```

OUTPUT

```
Enter first:Enter second:GCD is 12
```

Q3. sum to a target value

ANSWER

```
def find_ pairs(1st, target):
seen=set()
pairs=[]
for num in 1st
complement = target -num
if complement in seen:
pairs. append((complement, num))
seen. add(num)
return pairs
data=[int(x) for x in input("Enter numbers: ").split()]
t=int(input("Target: "))
result=find_ pairs(data, t)
print("pairs:" ,result)
```

Q4. Bitwise Operators

ANSWER

a=13, b=21, c=26, result=31
0b11111 0o37 0*1f
7 62 -32
a -> int
b->int
c->int
result->int

Q5. binary search

CODE

```
def rotated_binary_search(arr,target):
    low=0
    high=len(arr)-1
    while low≤high:
        mid=(low+high)// 2
        if arr[mid] == target:
            return mid
        if arr[low] ≤ arr[mid]:
            if arr[low] ≤ target <arr[mid]:
               high=mid-1
            else:
               low=mid+1
        else:
            if arr[mid]< target ≤ arr[high]:
                low = mid+1
            else:
                 high=mid-1
    return -1
data=[4,5,6,7,0,1,2]
t=7
result=rotated_binary_search(data,t)
print("found at:",result)
```

OUTPUT

```
found at: 3
```